

SHAUNAK NAGVENKAR

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Education

Vellore Institute of Technology, Vellore

B.Tech in Computer Science & Engineering

2021 - 2025

8.25/10 CGPA

Work Experience

Tata Consultancy Services (Full-time)

July 2025 – Present

Designation: Systems Engineer - Current Role: Big Data Engineer

- Engineered a cross-border gift card transaction pipeline using Apache Spark and Scala, enabling automated detection and routing of geographically invalid transactions (e.g., US-issued gift cards used in India) that were previously reconciled manually by merchants, reducing reconciliation overhead
- Designed and developed a reusable config-driven data validation library built on the Spark DataFrame API, migrating from Spark SQL to enable declarative multi-stage data transformations via configuration files rather than hardcoded SQL statements — reducing pipeline development time across the team and improving code maintainability, reusability, and readability. Additionally, leveraged internal AI tools to implement intelligent validation and anomaly detection mechanisms.

Remote Software Solutions (Internship)

August 2023 – September 2023

Data Analyst

- Designed and optimized relational database schemas using PostgreSQL to improve data organization and query efficiency.
- Developed and maintained SQL queries for data extraction, transformation, and analysis.
- Built interactive dashboards using Power BI to support business insights and decision-making.
- Collaborated with senior engineers to understand scalable system design and real-world software development practices.

Technical Skills

Languages: Python, Scala, JavaScript

Backend: FastAPI, Node.js

ML Frameworks and Technology: Agentic AI: LangChain(also LangGraph and LangSmith), CrewAI, Generative AI: LLMs(Transformer based models - DeepSeek, OpenAI, LLaMA), Ollama, Diffusion Models(learning), Variational AutoEncoder(learning), GAN's, MLOps(MLFlow) and LLMOps(LangChain), Deep and Statistical Machine Learning frameworks: Tensorflow, ScikitLearn

Clouds & Databases: AWS Cloud(EC2, S3, Lambda), PostgreSQL, MongoDB, Pinecone, ChromaDB

Data Tools and Technologies: Apache Spark, Apache Airflow, Pandas and Numpy, PowerBI

Developer Tools and Technologies: Docker, Postman, VS Code, GitHub

Projects

LLM Preference Classification Finetuning

[Source Code](#)

- Built a preference prediction system to determine user-preferred responses between competing LLM outputs using supervised finetuning.
- Implemented text preprocessing, structured prompt-response encoding, and trained transformer-based models (RoBERTa) optimized for multi-class classification (A/B/Tie).
- Designed an end-to-end ML pipeline including data cleaning, tokenization, model training, inference, and Kaggle submission workflow.
- Tech:** Python, PyTorch, HuggingFace Transformers, Tokenizers, Pandas, NumPy

Potato Plant Leaf Health Decision System Using Image-Based Analysis for Detecting Defects

[Source Code](#)

- Developed a deep learning-based system to detect plant diseases from leaf images and provide actionable health insights.
- Trained CNN architectures on labeled datasets to classify disease categories and improve early detection accuracy.
- Integrated preprocessing, augmentation, and model evaluation pipelines for robust performance across varying image conditions.
- Tech:** Python, TensorFlow/Keras, OpenCV, NumPy, Matplotlib

Task-Specific Personal Assistant using Agentic AI | In Progress

[Source Code](#)

- Built an agent-based AI assistant capable of executing task-specific workflows using LLM reasoning and tool integration.
- Implemented multi-step decision-making pipelines with dynamic prompt chaining and context management.
- Enabled automation of tasks such as query handling, summarization, and structured output generation.

- Integrated RAG based concepts to produce content more relevant to the conversatiozn.
- **Tech:** Python, CrewAI, Ollama, DeepSeek, FastAPI, ChromaDB

Invoice Summary Generator using AI Document Processing

[Source Code](#)

- Designed an AI-powered system to extract and summarize key information from invoice documents.
- Implemented OCR-based text extraction followed by automating financial document processing to reduce manual effort and improve accuracy in data extraction.
- **Tech:** React, JavaScript, Mindee OCR Image Parsing API, FastAPI

Restaurant Management System

[Source Code](#)

- Engineered a full-stack restaurant management system with a strong focus on database design, enabling real-time handling of orders, inventory, and customer data.
- Designed highly normalized relational schemas in PostgreSQL, leveraging indexing, joins, and query optimization techniques to ensure efficient data retrieval and transactional consistency.
- **Tech:** PostgreSQL, ReactJS, NodeJS, ExpressJS

Airport Management Information System — Airport Authority of India, Kalaburagi

[Source Code](#)

- Developed a web-based internal management platform for the **Airport Authority of India (AAI)**, streamlining operations across Fire, City-Side, Terminal, Airside, Communication, Navigation and Surveillance (CNS), and Electrical departments.
- Designed and implemented a **User Management** module enabling department administrators to manage staff profiles, monitor attendance, and oversee task assignments through a centralized dashboard.
- Built a **Daily Task Management** system with department-specific workflows, allowing administrators to assign, monitor, update, and verify operational tasks across multiple departments.
- Implemented an internal **Messaging** module and a **BCAS Compliance Dashboard** centralizing safety inspections, emergency drills, contingency plans, regulatory updates, and other compliance records into a unified interface.

AI-Powered Internal Knowledge Assistant

[Source Code](#)

- Designed an AI-powered system to extract and summarize key information from invoice documents.
- Implemented OCR-based text extraction followed by automating financial document processing to reduce manual effort and improve accuracy in data extraction.
- **Tech:** React, JavaScript, Mindee OCR Image Parsing API, FastAPI

Taxi Demand Prediction Pipeline

[Source Code](#)

- Built an end-to-end MLOps pipeline to automate NYC taxi demand prediction, integrating data ingestion, preprocessing, feature engineering, model training, evaluation, and artifact versioning.
- Orchestrated monthly workflows using Apache Airflow and implemented CI/CD with GitHub Actions for automated model retraining, validation, and deployment-ready pipelines.
- **Tech:** Python, Apache Airflow, Scikit-learn, Pandas, MLflow, MinIO, GitHub Actions, Docker, PostgreSQL

Research, Publications and Patents

Generative AI-Enhanced Data Marketplaces: Algorithms and Platforms for Secure Data Exchange [Published](#)

- **Patent Application pending** - Application No: 202541123233
- **Published in IEEE Publication**
- Presented at the **4th IEEE International Conference on Sensors and Related Networks SENNET-2025 Special Focus on Digital Healthcare** on the 26th of July, 2025.
- Built a web marketplace enabling users to generate, share, and access AI-enhanced datasets, automating dataset creation with generative models to improve data diversity and ML project quality.
- Tech Stack: JavaScript, TensorFlow, ReactJS, NodeJS, Pinecone

Comparative Analysis of Fine-Tuned Convolutional Models in Detecting Face Masks

[Source Code](#)

- Analyzed VGG, ResNet, Inception, and MobileNet for face-mask detection, focusing on fine-tuning, data augmentation, and optimization for public health applications.

Certifications

- **Mastering GPU Parallel Programming with CUDA** *Udemy* Jul 2026
- **Introduction to Model Context Protocol (MCP)** *Anthropic* May 2026
- **Apache Spark Programming in Scala** *Udemy* Apr 2026
- **Amazon S3 Mastery** *Udemy* Apr 2026
- **Generative AI Fundamentals** *IBM* Jun 2025
- **Supervised Machine Learning: Regression and Classification** *DeepLearning.AI (Coursera)* Sep 2023

Competitions and Participation

TCS AI Fridays (TCS Internal Hackathon)

- **Agentic AI Powered Fault Detection System for Electrical Appliances**
- Built a vector database from appliance manuals using ChromaDB and implemented Retrieval-Augmented Generation (RAG) to extract operational thresholds and parameter ranges.
- Compared real-time sensor readings against retrieved reference values to detect anomalies and identify potential faults.
- Designed an agentic workflow using LangChain and LangGraph to automate reasoning, validation, and fault diagnosis using LLMs.
- Stack used: **LangChain, LangGraph, ChromaDB (Vector DB, RAG), DeepSeek (LLM)**